

Vis-FAB:

Visual Analytics Framework for Wafer Failure Analysis in Multistage Fabrication Process

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Problem Statement

- 1. Process TBs/day of data
- 2. Alt-tab across tasks
- 3. Explain what, why, and what-if

Research Question

Can we merge AI predictions and multiscale wafer data into a single, explainable, end-to-end visual analytics workflow?

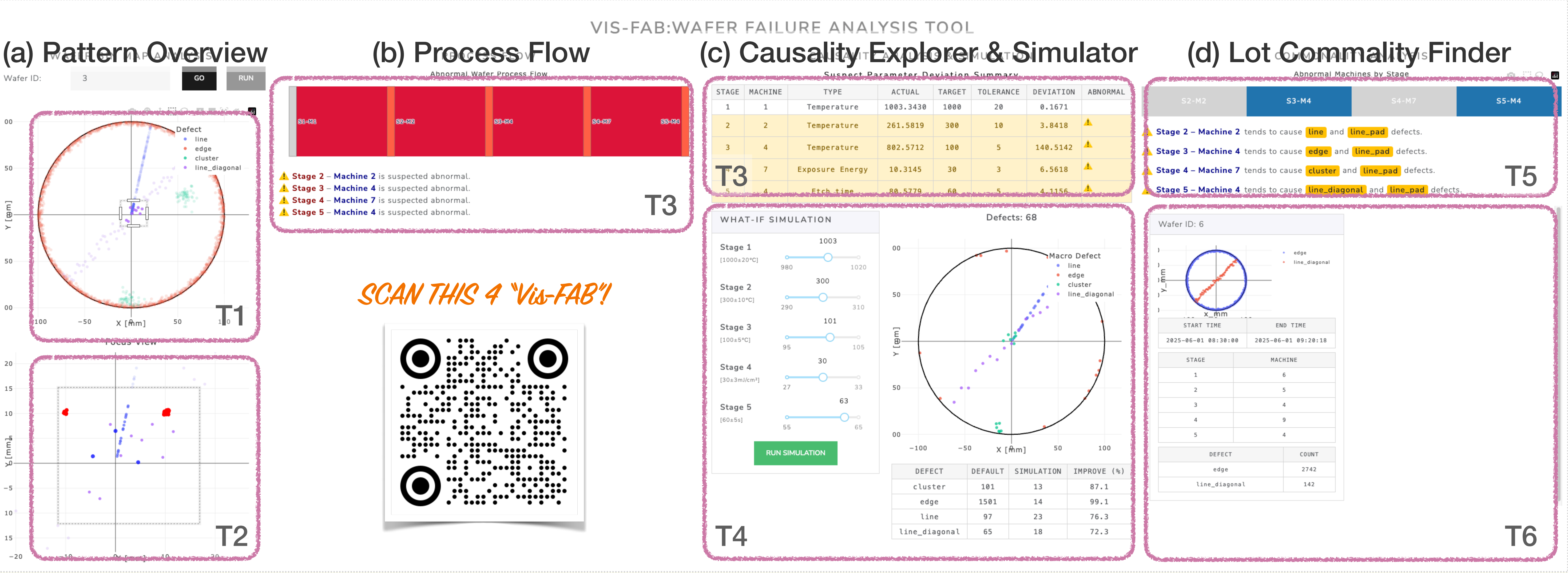
Design Goals

- G1 End-to-end Integration

To provide a single, continuous workflow that covers the entire failure analysis loop, supporting engineers' seamless navigation across *overview* → *zoom* → *detail* → *context*
- G2 Explainable AI

To transform black-box model outputs into domain-specific, interactive visual explanation for engineers to understand *what* the model predicts, *why* an anomaly is flagged, and *how* a parameter change would *fix* it

User Interface



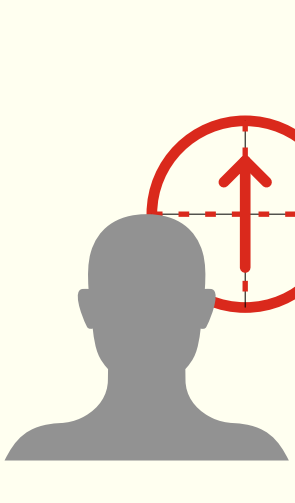
Task → View	(a) Pattern Overview	(b) Process Flow	(c) Causality Explorer & Simulator	(d) Lot Commonality Finder
T1 Identify multiclass defect on a WBM	✓			
T2 Integrate multiscale defect with a single figure - macro defect (WBM) & micro defect (chip)	✓			
T3 Determine which stage & machine is most likely abnormal		✓	✓	
T4 Adjust process parameters and observe predicted effects in-situ			✓	
T5 Analyze which stage & machine typically generates what defect pattern				✓
T6 Find other lots processed with the same machine subset and conduct a comparative failure analysis				✓

User Feedback



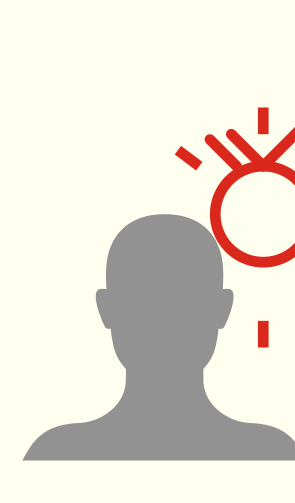
Quality Assurance Engineer says

- Super traceability
- Strong model exploitability
- Good integration of features



Yield Enhancement Engineer says

- Rapid cluster spotting
- Instant defect-rate predictions
- Request for long-term trend



Unit Process Engineer says

- Fast machine-level analysis
- Useful when tuning recipes
- Desire for multi-parameter interpretation